

UNIT-4 | Lecture 3

Normal Forms — 1NF, 2NF & 3NF

Coverage: Hour 3

Learning Objectives

Understand normalization and anomalies.

Identify 1NF, 2NF, and 3NF requirements.

Normalization

Normalization organizes relations to reduce redundancy and prevent insertion, deletion, and update anomalies.

1NF — First Normal Form

A relation is in **1NF** when attribute values are atomic and there are no repeating groups or multiple values stored in a single cell.

Example violation: Student(Student_ID, Name, Phone_Numbers) when Phone_Numbers contains several phone numbers in one cell.

2NF — Second Normal Form

A relation is in **2NF** if it is in 1NF and every non-prime attribute is fully functionally dependent on every candidate key.

2NF mainly matters when a candidate key is composite. It removes **partial dependency**.

Example: ENROLL(Student_ID, Course_ID, Student_Name, Grade). If (Student_ID, Course_ID) → Grade but Student_ID → Student_Name, Student_Name has a partial dependency.

3NF — Third Normal Form

A relation is in **3NF** if it is in 2NF and every non-trivial FD $X \rightarrow A$ has either X as a superkey or A as a prime attribute.

It removes problematic **transitive dependencies** involving non-prime attributes.

Quick Comparison

Normal Form	Main Requirement
1NF	Atomic values; no repeating groups
2NF	1NF + no partial dependency
3NF	2NF + every non-trivial $X \rightarrow A$ has X as superkey or A prime

Key Takeaway

1NF handles atomicity, **2NF** removes partial dependency, and **3NF** removes problematic transitive dependency.